

FAMILIES OF MULTIVARIATE DISTRIBUTIONS HAVING PROPERTIES USUALLY ASSOCIATED WITH THE WISHART DISTRIBUTION

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1. Introduction

In this article we shall consider some general families of multivariate probability density functions (p.d.f.'s) with the same properties. We shall however, restrict ourselves to the following two properties namely the Wishart and the ratio property.

It is well known that the Wishart distribution has the property that

$$[M(\frac{\theta}{\alpha}, \Sigma^{-1}, n)]^{\alpha} = M(\theta; \alpha \Sigma^{-1}, \alpha n)$$

where $M(\cdot)$ denotes the common moment generating function (m.g.f.) (Anderson (1958)) and α the size of a sample of random matrices. From the above m.g.f. it thus follows, by using the inverse Laplace transform (Constantine (1963)), that the distribution of the sample mean for a simple random sample from a Wishart distribution, again has a Wishart distribution with the parameters scaled by the sample size. In section 2 we shall point out that this is not a property unique to the Wishart, but that one can find a family of p.d.f.'s other than the Wishart with this property.

If $X_1 : p \times p$ and $X_2 : p \times p$ are independent random matrices both drawn from a Wishart distribution, then it is known that the distribution of $X_2^{-1/2} X_1 X_2^{-1/2}$ has a Beta type II distribution (Olkin and Rubin (1964)). The question now arises that if $X_1 : p \times p$ and $X_2 : p \times p$ are independent identically distributed matrices drawn from a certain distribution and $X_2^{-1/2} X_1 X_2^{-1/2}$ has a Beta type II distribution, does it necessarily mean that X_1 and X_2 come from a Wishart distribution? We shall show in section 3 that this need not be the case, by considering a p.d.f. (Van der Merwe and Roux (1974)) different from the Wishart for which the ratio turns out to be a Beta type II distribution.

2. The Wishart Property

The Wishart property may in general be defined as

$$[M(\frac{\theta}{\alpha}; n_1, \dots, n_k)]^\alpha = M(\theta; \alpha n_1, \dots, \alpha n_k),$$

where $n_1, \dots, n_k$ are parameters

Theorem 2.1

If $X : p \times p$ has the p.d.f.

$$\begin{aligned} f(X; P, B, D, \nu, m) &= \frac{|P-B|^{2\nu-2m}}{\Gamma_p(2\nu-2m) {}_1F_1(2\nu; (D-B)(P-B)^{-1})} \\ &\cdot e^{-\text{tr } X(P-B)} |X|^{2\nu-2m-\frac{1}{2}(p+1)} \\ &\cdot {}_1F_1(2\nu; 2\nu-2m; (D-B)X) \end{aligned}$$

for $X > 0$, $\text{Re}(\nu-m) > \frac{1}{2}(p-1)$ and $f(X; P, B, D, \nu, m) = 0$ elsewhere, then the sample mean has the distribution $f(X; \alpha P, \alpha B, \alpha D, \alpha \nu, \alpha m)$ where $P : p \times p$, $B : p \times p$ and $D : p \times p$ are positive definite matrix parameters. The hypergeometric function ${}_1F_q(\cdot)$ is defined by Constantine (1963).

Proof

From Constantine (1963) we have

$$\begin{aligned} \int_{S > 0} e^{-\text{tr } S(P-B)} |S|^{2\nu-2m-\frac{1}{2}(p+1)} {}_1F_1(2\nu; 2\nu-2m; (D-B)S) dS \\ = \Gamma_p(2\nu-2m) |P-B|^{2m} |P-D|^{-2\nu} \end{aligned}$$

Hence the m.g.f. is

$$M(T; P, B, D, \nu, m) = |P-B|^{-2m} |P-D|^{2\nu} |P-B-T|^{2m} |P-D-T|^{-2\nu}.$$

Now it is easily seen that

$$[M(\frac{T}{\alpha}; P, B, D, \nu, m)]^\alpha = M(T; \alpha P, \alpha B, \alpha D, \alpha \nu, \alpha m).$$

This m.g.f. satisfies the conditions (Constantine (1963)) for the existence of the inverse Laplace transform. From the uniqueness of this transform the theorem follows.

Theorem 2.2

If $X : p \times p$ has the p.d.f.

$$f(X; P, B, D, t) = \frac{|P|^t}{\Gamma_p(t)} e^{-\text{tr } PB^{-1/2}DB^{-1/2}} e^{-\text{tr } PX} |X|^{t-1/2(p+1)} \\ \cdot {}_1F_0(-t + 1/2(p+1); -B^{-1/2}DB^{-1/2}X^{-1})$$

for $X > 0$, $\text{Re}(t) > k/2 + 1/2(p-1)$ and $f(X; P, B, D, t) = 0$ elsewhere, then the sample mean has the distribution $f(X; \alpha P, \alpha B, \alpha D, \alpha t)$.

Proof

From Khatri (1966) we have,

$$\int_{S > 0} e^{-\text{tr } PS} |S|^{t-1/2(p+1)} C_K(B^{-1/2}DB^{-1/2}S^{-1}) dS \\ = \Gamma_p(t, -K) |P|^{-t} C_K(PB^{-1/2}DB^{-1/2}).$$

Using a result similar to the above in Constantine (1966) we obtain

$$\Gamma_p(t, -K) = \frac{(-1)^k \Gamma_p(t)}{(-t + 1/2(p+1))_K}.$$

Hence the m.g.f. is

$$M(T; P, B, D, t) = |P|^t |P-T|^{-t} e^{-\text{tr } TB^{-1/2}DB^{-1/2}}$$

and

$$[M(\frac{T}{\alpha}; P, B, D, t)]^\alpha = M(T; \alpha P, \alpha B, \alpha D, \alpha t).$$

This completes the proof.

In the above theorems, positivity of the functions is assumed since this can be easily achieved by selecting the parameters properly.

3. The Ratio Property

Let $X : p \times p$ be a random matrix with p.d.f. given by

$$(3.1) \quad f(X) = \begin{cases} C|X|^{\frac{1}{2}\gamma - \frac{1}{4}(p+1)} {}_1F_1(a + \frac{1}{2}\gamma + \frac{1}{4}(p+1); \gamma + \frac{1}{2}(p+1); -X), & X > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where

$$C = \frac{\Gamma_p(a + \frac{1}{2}\gamma + \frac{1}{4}(p+1))}{\Gamma_p(a) \Gamma_p(\gamma + \frac{1}{2}(p+1))}.$$

The parameters a and γ are restricted to take values for which $f(X)$ is positive.

Lemma 3.1 (Van der Merwe and Roux (1974))

$$\int_{X > 0} f(X) dX = 1.$$

Definition 3.1

The Hankel transform of a function $f(R)$ is defined by (Herz (1955))

$$g(\Lambda) = \int_{R > 0} A_\gamma(\Lambda R) |R|^\gamma f(R) dR$$

and the inverse Hankel transform as

$$f(N) = \int_{\Lambda > 0} A_\gamma(\Lambda N) |\Lambda|^\gamma g(\Lambda) d\Lambda$$

whenever the integrals are absolutely convergent, γ is a real number greater than -1 and the generalised Bessel function

$$A_\gamma(\Lambda R) = \frac{1}{\Gamma_p(\gamma + \frac{1}{2}(p+1))} \sum_{k=0}^{\infty} \frac{C_K(-\Lambda R)}{K (\gamma + \frac{1}{2}(p+1))_K k!}$$

with $C_K(R)$ defined by James (1964)

Theorem 3.1

If $\tilde{g}(\Lambda)$ and $\tilde{f}(\Lambda)$ are the Hankel transforms of $g(R)$ and $f(R)$ respectively, then

$$\int_{\Lambda > 0} |\Lambda|^\gamma \tilde{g}(\Lambda) \tilde{f}(\Lambda) d\Lambda = \int_{R > 0} |R|^\gamma g(R) f(R) dR$$

provided the integrals involved converge absolutely.

Proof.

From definition 3.1 it follows that

$$\int_{\Lambda > 0} |\Lambda|^\gamma \tilde{g}(\Lambda) \tilde{f}(\Lambda) d\Lambda = \int_{\Lambda > 0} |\Lambda|^\gamma \tilde{g}(\Lambda) \int_{R > 0} A_\gamma(\Lambda R) |R|^\gamma f(R) dR d\Lambda.$$

Changing the order of integration, which is justifiable, and using definition 3.1, we get

$$\int_{\Lambda > 0} |\Lambda|^\gamma \tilde{g}(\Lambda) \tilde{f}(\Lambda) d\Lambda = \int_{R > 0} |R|^\gamma f(R) g(R) dR.$$

This proves the theorem.

Lemma 3.2

The Hankel transform of

$$\frac{1}{\Gamma_p(a)} |R|^{a-\frac{1}{2}\gamma-\frac{1}{4}(p+1)} e^{-\text{tr } Y^{-1}R}$$

is given by :

$$\int_{R > 0} |R|^\gamma A_\gamma(\Lambda R) \frac{1}{\Gamma_p(a)} |R|^{a-\frac{1}{2}\gamma-\frac{1}{4}(p+1)} e^{-\text{tr } Y^{-1}R} dR$$

$$= \frac{\Gamma_p(a + \frac{1}{2}\gamma + \frac{1}{4}(p+1))}{\Gamma_p(a) \Gamma_p(\gamma + \frac{1}{2}(p+1))} |Y|^{a+\frac{1}{2}\gamma+\frac{1}{4}(p+1)}$$

$${}_1F_1(a + \frac{1}{2}\gamma + \frac{1}{4}(p+1); \gamma + \frac{1}{2}(p+1); -\Lambda Y).$$

Proof

By using an integral similar to the one which proves Theorem 2.1 and the definition of $A_\gamma(\Lambda R)$, the lemma follows.

Theorem 3.2

$X_1 : p \times p$ and $X_2 : p \times p$ are independent random matrices with p.d.f.'s given by $f_{a_i}(X_i)$ for $i=1,2$ where $f_{a_i}(X_i)$ is defined in (3.1). Then $X_2^{-\frac{1}{2}} X_1 X_2^{-\frac{1}{2}}$ has a Beta type II

distribution for all values of γ such that $f_{a_i}(X)$ exists.

Proof

The joint p.d.f. of X_1 and X_2 is given by

$$\begin{aligned} f(X_1, X_2) &= \frac{\Gamma_p(a_1 + \frac{1}{2}\gamma + \frac{1}{4}(p+1)) \Gamma_p(a_2 + \frac{1}{2}\gamma + \frac{1}{4}(p+1))}{\Gamma_p(\gamma + \frac{1}{2}(p+1)) \Gamma_p(a_1) \Gamma_p(\gamma + \frac{1}{2}(p+1)) \Gamma_p(a_2)} \\ &\quad \cdot |X_1|^{\frac{1}{2}\gamma - \frac{1}{4}(p+1)} |X_2|^{\frac{1}{2}\gamma - \frac{1}{4}(p+1)} \\ &\quad \cdot {}_1F_1(a_1 + \frac{1}{2}\gamma + \frac{1}{4}(p+1); \gamma + \frac{1}{2}(p+1); -X_1) \\ &\quad \cdot {}_1F_1(a_2 + \frac{1}{2}\gamma + \frac{1}{4}(p+1); \gamma + \frac{1}{2}(p+1); -X_2). \end{aligned}$$

Let $Y = X_2^{-\frac{1}{2}} X_1 X_2^{-\frac{1}{2}}$ with $J(X_1 \rightarrow Y) = |X_2|^{\frac{1}{2}(p+1)}$ and integrating over X_2 we obtain

$$\begin{aligned} f(Y) &= \frac{\Gamma_p(a_1 + \frac{1}{2}\gamma + \frac{1}{4}(p+1)) \Gamma_p(a_2 + \frac{1}{2}\gamma + \frac{1}{4}(p+1))}{\Gamma_p(\gamma + \frac{1}{2}(p+1)) \Gamma_p(a_1) \Gamma_p(\gamma + \frac{1}{2}(p+1)) \Gamma_p(a_2)} \\ &\quad \cdot |Y|^{\frac{1}{2}\gamma - \frac{1}{4}(p+1)} \\ &\quad \cdot \int_{X_2 > 0} |X_2|^\gamma {}_1F_1(a_1 + \frac{1}{2}\gamma + \frac{1}{4}(p+1); \gamma + \frac{1}{2}(p+1); -YX_2) \\ &\quad \cdot {}_1F_1(a_2 + \frac{1}{2}\gamma + \frac{1}{4}(p+1); \gamma + \frac{1}{2}(p+1); -X_2) dX_2. \end{aligned}$$

Using Theorem 3.1 and Lemma 3.2 we have

$$\begin{aligned} f(Y) &= \frac{1}{\Gamma_p(a_1) \Gamma_p(a_2)} |Y|^{-a_1 - \frac{1}{2}(p+1)} \\ &\quad \cdot \int_{R > 0} |R|^{a_1 + a_2 - \frac{1}{2}(p+1)} e^{-\text{tr } R(1+Y^{-1})} dR. \end{aligned}$$

From Herz (1955) it follows that

$$f(Y) = \frac{\Gamma_p(a_1 + a_2)}{\Gamma_p(a_1) \Gamma_p(a_2)} |Y|^{a_2 - \frac{1}{2}(p+1)} |I + Y|^{-(a_1 + a_2)}$$

which proves the theorem.

Thus one can evaluate certain infinite integrals with the help of theorem 3.1. Theorem 3.2 however, shows that the ratio of two matrices may have a Beta type II distribution without the components necessarily being Wisharts.

Remark

These properties also hold for univariate p.d.f.'s. and have been considered by several authors.

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